

Math 115A Spring 2026 Discussion 13 Handout

Ben Slater
UCLA Mathematics Department

May 12, 2026

1. Introduction

Today we'll talk about more properties of linear transformations: kernels and images (and ranks), and then eigenvalues and eigenvectors.

2. Kernels and Images

Let's start with the definitions:

Definition: Let $T : V \rightarrow W$ be a linear transformation. The kernel of T , written $\ker(T) \subset V$, is the set of $v \in V$ such that $Tv = 0$. The image of T , written $\operatorname{im}(T) \subset W$, is the set of $w \in W$ such that for some $v \in V$, $Tv = w$.

Exercise: Show that $\ker(T)$ is a subspace of V , and that $\operatorname{im}(T)$ is a subspace of W .

Why define these terms at all? Well, for one thing, they give a convenient language to describe results on injectivity and surjectivity.

Exercise: Let $T : V \rightarrow W$ be a linear transformation such that $\ker(T) = \{0\}$. Then T is injective. (I gave this as an exercise on Thursday of Week 5, although I didn't use quite this language then; it was also shown in class. But it's worth working out for yourself!)

Exercise: Let $T : V \rightarrow W$ be a linear transformation such that $\ker(T) = \{0\}$; show that $V \cong \operatorname{im}(T)$.

Easy Exercise: Let $T : V \rightarrow W$ be a linear transformation. such that $\operatorname{im}(T) = \{0\}$. Show that $V \cong \ker(T)$.

Challenging Exercise, for those who have worked through the “bonus material” on quotient spaces: Let $T : V \rightarrow W$ be a surjective linear transformation. Show that $W \cong V/\ker(T)$.

But there is a better reason to study kernels and images. The original purpose of linear algebra was to help us solve systems of equations like

$$\begin{cases} 2x + 4y - z &= 8 \\ 6x - y + 3z &= 5 \end{cases}$$

When an equation like this has a solution, it's not hard to arrive at it by adding and subtracting the equations from each other - at least when the number of equations and variables is small. But this is not tractable for larger problems - and doesn't even make sense when dealing with infinite-dimensional problems (more on this in a bit).

With our more abstract perspective, we can not only develop more systematic approaches to solving systems of linear equations, but also begin asking (and answering) more sophisticated questions. When a mathematician sees an equation, the first two questions they ask are:

- (a) Does this equation have a solution?
- (b) If so, how many solutions does it have?

Only after determining that a solution exists (and after figuring out how many solutions they should look for) does a mathematician set about trying to solve an equation.

To answer the mathematician's questions, we want to cast the question in more abstract terms. What we have in the above example is a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. An element of the domain is given by a column vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$, and $T\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is given (in the standard basis of $\mathbb{R}^3$) by computing

$$\begin{pmatrix} 2 & 4 & -1 \\ 6 & -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Asking when a solution to our original system of equations exists amounts to asking when the column vector $\begin{pmatrix} 8 \\ 5 \end{pmatrix}$ is in the image of T . (Note that simply rephrasing the question does not make it any easier! In fact, a priori, computing the entire image of T is much more difficult than solving a single system of equations. On the other hand, the number of solutions to a system of equations like this turns out to be related to the kernel of T .)

Exercise: Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the linear transformation representing a system of equations; that is, we're looking for solutions to the equation $Tv = w$. Show that if v_0 is a solution to this equation, i.e. $Tv_0 = w$, and $u \in \ker(T)$, then $T(v_0 + u) = w$. In fact, show that *every* solution to the equation is of the form $T(v_0 + u)$, for v_0 fixed and $u \in \ker(T)$.

That is, a system of linear equations will always have "a vector space's worth" of solutions (a point, or a line, or a plane, or etc.). Technically, this is not really a vector space but rather what is known as an "affine space" - it is geometrically/set theoretically equivalent to a vector space, but is generally not a vector space.

Exercise: Suppose $Tv = w$ is an equation like the one above. Show that if $w \neq 0$, then the set $U = \{v \in V : Tv = w\}$ is not a subspace of V . (In fact, it's not even close - there are at least 3 reasons why not).

But how can we get at the kernel and image of a vector space? The key result is what, in this course, is called the Dimension Theorem:

Theorem: Let $T : V \rightarrow W$ be a linear transformation between finite-dimensional vector spaces. Then $\dim(\ker(T)) + \dim(\text{im}(T)) = \dim(V)$.

Remark: $\dim(\ker(T))$ is often called the *nullity* of T , and $\dim(\text{im}(T))$ is often called the *rank* of T . For this reason, the above theorem is also called the *rank-nullity theorem*.

Exercise: Without looking at the course notes, prove this.

Exercise: Let $T : V \rightarrow W$ be a linear transformation, for $\dim(V), \dim(W) < \infty$. Show that the $\text{rank}(T) < \dim(V)$ and $\text{rank}(T) < \dim(W)$.

Thus, a map $\mathbb{R}^2 \rightarrow \mathbb{R}^4$ can never be surjective; its rank is at most 2. On the other hand, a map $\mathbb{R}^4 \rightarrow \mathbb{R}^2$ can never be injective: its rank is at most 2, and so its nullity (dimension of the kernel) has to be at least 2.

3. **Determinants** We've seen that (finite-dimensional) vector spaces over a field are entirely classified up to dimension. You might wonder if we can say something similar about linear transformations - or to be more precise, about matrices.

Question: Let M, N be two $n \times n$ matrices. Suppose $\text{rank}(M) = \text{rank}(N)$. Is there an invertible matrix P such that $P^{-1}MP = N$?

That is, if two matrices have the same rank, do they represent the same linear transformation in different coordinates?

The answer is no: linear transformations have much more structure than just their rank, and therefore have a much richer theory. This is easiest to make sense of for maps from a vector space V to itself. So consider a linear transformation $T : V \rightarrow V$. The most important property of T , beyond its rank, is its set of eigenvalues and eigenvectors.

Let's recall the definition:

Definition: Let V be a vector space over a field $\mathbb{F}$, and $T : V \rightarrow V$ be a linear transformation. An *eigenvector* for T is a (nonzero) vector $v \in V$ such that $Tv = \lambda v$, for some $\lambda \in \mathbb{F}$. An *eigenvalue* for T is a scalar $\lambda \in \mathbb{F}$ such that there is an eigenvector v of T with $Tv = \lambda v$.

Note that a single eigenvalue can correspond to more than one eigenvector; on the other hand, each eigenvector corresponds to just one eigenvalue.

Exercise: Let $T : V \rightarrow V$ be a linear transformation and λ an eigenvalue for T . Let $E_\lambda(T)$ denote the set of vectors $v \in V$ such that $Tv = \lambda v$ (i.e. the set of eigenvectors for the eigenvalue λ). Show that $E_\lambda(T)$ is a subspace of V .

Another way of defining $E_\lambda(T)$ is as $\ker(\lambda I_V - T)$, where I is the identity map on V ; this makes the above exercise obvious in retrospect.

The set of eigenvalues of a linear transformation T is also denoted $\sigma(T)$, called the *spectrum* of T .¹ The spectrum is what we might call an “invariant” of a linear transformation, for the following reason:

Exercise: Let $T : V \rightarrow V$ be a linear transformation and $P : V \rightarrow V$ be an invertible linear transformation. Show that $\sigma(T) = \sigma(P^{-1}TP)$.

Exercise: For any nonzero scalar μ , find a relationship between $\sigma(T)$ and $\sigma(\mu T)$.

Exercise: By thinking geometrically, describe a family of maps $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\sigma(T) = \emptyset$.

The best way to find $\sigma(T)$ is to look for solutions to the characteristic polynomial $\det(xI - T) = 0$; if this holds for some x , then $xI - T$ is not invertible, hence not injective; an element of the kernel is an eigenvector for x . Since $\sigma(T) = \sigma(P^{-1}TP)$ for any invertible P , we can compute the spectrum of T by picking any matrix representation of T . Once we have a matrix representation M , we consider $\det(xI - M)$, and solve.

Exercise: Find $\sigma(T)$ if T has a matrix representation

$$\begin{pmatrix} 1 & 5 & 3 \\ 2 & 4 & -1 \\ 0 & 3 & 2 \end{pmatrix}$$

Note that this technique is only really computable for small matrices. The degree of the characteristic polynomial is the dimension of V ; in general, there is no systematic way to solve polynomials of degree greater than 4. So for large matrices, the best one can get in practice is a good approximation to the eigenvalues. Still, the characteristic determinant is extremely useful for proving theoretical results.

Exercise: Suppose V is a finite-dimensional vector space and $T : V \rightarrow V$ is a map. Show that $|\sigma(T)| \leq \dim(V)$.

Exercise: Suppose V is a finitely-dimensional vector space over $\mathbb{C}$, and $T : V \rightarrow V$ is a $\mathbb{C}$ -linear map. Show that T has at least one eigenvector.

Remark: You might be tempted to take the above exercise even further, and say that T has exactly $\dim(V)$ linearly independent eigenvectors. For reasons we might see later in the quarter, the truth is a bit subtler than that.

Definition: A linear transformation $T : V \rightarrow V$ is called diagonalizable if there is a basis $B = \{v_1, \dots, v_n\}$ of V , called an eigenbasis such that each v_i is an eigenvector.

¹Spectrum seems to be every mathematician’s favorite word. I can think of at least three completely unrelated examples off the top of my head, plus a fourth, also unrelated, use of the word “spectral.” The linear algebra concept seems to me to be the best fit for the term.

Exercise: Show that in particular, if T is diagonalizable and B is an eigenbasis, then the matrix for T with respect to B is diagonal. This explains the name.

Exercise: The trace of a matrix M , denoted $\text{tr}(M)$ is the sum of the entries along its diagonal. Suppose $T : V \rightarrow V$ is diagonalizable, and let M be a matrix representation of T . Show that $\text{tr}(M)$ is equal to the sum of the eigenvalues of T , and that $\det(M)$ is equal to the product of the eigenvalues of T . Conclude that the trace of a diagonalizable matrix M is the same as the trace of $P^{-1}MP$ for any invertible matrix P .

Remark: Actually, it is always true that $\text{tr}(M) = \text{tr}(P^{-1}MP)$; you can use the same argument as in the above exercise, but you need to be able to define the characteristic polynomial of a linear transformation T , or else show that M and $P^{-1}MP$ have the same characteristic polynomial. This is a bit tricky; the easiest way I can think of uses ideas about invertible matrices that I don't believe we've covered.

The trace looks a bit silly compared to the sophistication of the determinant. Bizarrely, the trace turns out to be the more abstractly meaningful invariant, although the examples are a bit beyond the scope of this class. (If you ask during office hours, I can try to give a medium-length explanation).

4. **Bonus topic: the infinite dimensional case** For a more abstract example, let's look at a couple of infinite-dimensional vector spaces. Recall that $C^1([0, 1])$ is the space of "differentiable functions on $[0, 1]$ with continuous derivatives" and $C^0([0, 1])$ is the space of "continuous functions on $[0, 1]$." Recall, as well, that the derivative gives a linear transformation $\frac{d}{dx} : C^1([0, 1]) \rightarrow C^0([0, 1])$.

Exercise: What is $\ker(\frac{d}{dx})$?

When you solve an indefinite integral, that is, when you try to compute $\int f(x)dx$, you are actually doing a linear algebra problem! Namely, you are looking for a solution to the equation $\frac{d}{dx}g = f$. Once you've found one solution g_0 to this integral, you can find other "anti-derivatives" of f by adding elements of $\ker(\frac{d}{dx})$; this is, conceptually, where the $+c$ comes from.

Upgrading to multivariate calculus gives a more sophisticated example. Recall that for a function $\mathbb{R}^3 \rightarrow \mathbb{R}$ we can take its gradient ∇f , which is a vector field on $\mathbb{R}^3$. Once we have a vector field X , we can take its curl $\nabla \times X$ to get another vector field. There is a useful identity:

$$\nabla \times (\nabla f) = 0$$

In other words, the curl of a gradient is zero. Letting V be the space of differentiable functions on $\mathbb{R}^3$ and W the space of vector fields, we see that we have linear maps

$$V \xrightarrow{\text{grad}} W \xrightarrow{\text{curl}} W$$

Suppose we are given a vector field X and want to find a function f such that $\text{grad}(f) = X$. At the very least, since the curl of the gradient of a function is zero, we must have $X \in \ker(\text{curl})$. As it turns out, for vector fields defined on all of $\mathbb{R}^3$ $\ker(\text{curl}) = \text{im}(\text{grad})$; so, as long as X has zero curl, it is the gradient of some function f . If we remove the origin, this is no longer true: there is the vector field

$$V(x, y) = \left\langle \frac{-y}{x^2 + y^2}, \frac{-x}{x^2 + y^2} \right\rangle$$

As it turns out, the kernel of curl is exactly one dimension larger than the image of grad (in the language of quotient spaces: $\ker(\text{curl})/\text{im}(\text{grad}) \cong \mathbb{R}$, generated by the vector field $V(x, y)$ above. This is a first example of a tool called De Rham cohomology, which associates to a geometric space a vector field which measures the failure of solving this “gradient problem” (and other, related problems).